

Radiographic periapical healing associated with root treated teeth accessed through existing crowns: a historical controlled cohort study

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INTRODUCTION

Non-surgical Root canal treatment (NSRCT) aims to prevent or treat apical periodontitis [1] with the ultimate post-operative goal of normal periradicular tissue architecture. A range of pre-operative, intra-operative and post-operative prognostic factors influence periapical healing [2]. Of these, the quality of the post-treatment restoration has a decisive influence on long-term periapical healing, presumably through aiding control of the treated root canal system environment [3]. A well-designed, fabricated and delivered crown after NSRCT may also reduce uncontrolled mechanical stresses and the risk of catastrophic tooth structure failure [4] and thus improve the long-term survival of the tooth-full coverage restoration assembly.

It is estimated that millions of conventional crowns or fixed partial dentures (FPD) are placed every year [5,6], with or without prior root canal treatment. The placement of such restorations requires sufficient tooth structure sacrifice to create space for an adequate bulk of restorative material for optimal strength, contour and aesthetics. In a proportion of teeth with vital pulps, this may lead to severe and irreversible pulpal injury [7,8], and the consequent need for root canal treatment [9,10]. The proportion of restored teeth requiring root canal treatment may be related to the type of restoration and extent of occlusal support it provides. For example, 3% of teeth prepared for single crowns needed root canal treatment (at an average of 6 years post-operatively), compared to 11% of fixed partial denture abutments (at an average of 8 years post-operatively), and 10-23% of teeth supporting extensive fixed partial dentures [8,11]. Root canal treatment, in the form of revision, may also be required on crowned and previously root-treated teeth with persistent or new periapical disease [2,12].

1 The dentist confronted with the prospect of managing pulpal or periapical
2 disease on crowned teeth, faces the dilemma of whether to first remove the crown
3 for safer access or to cut an access through it [13]. The conscientious dentist will
4 be compelled to weigh several arguments for or against each approach, in the
5 process of deciding.
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11 Forceful prosthetic crown removal would risk damage to the tooth and
12 crown, whilst cutting the crown off would commit the patient to the expense of a
13 new restoration, albeit with tooth-structure saving. Access through a prosthetic
14 crown, meanwhile, may risk rendering the core and crown unstable through
15 unintentional weakening of carefully placed retention features [14]. It may also
16 lead to loss of orientation by the operator, leading to a mal-aligned access cavity
17 with unnecessary dentine removal, coupled with the ignominy of difficulty in
18 locating the canals [15].
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31 The argument for removal of crown or FPD prior to root canal treatment
32 was given a decisive edge by the notion and evidence that pulp necrosis may be
33 preceded by an underlying cause (leakage, caries or cracks) that merits prior
34 correction [16]. Added to this is the need to assess remaining tooth structure to
35 determine future predictable restorability. The counter-argument is that such a
36 draconian policy is prone to unnecessary sacrifice of sound restorations and
37 increased time and financial cost for the patient. With or without agonising about
38 such difficult decisions, dentists and patients frequently choose to keep an
39 existing extra-coronal restoration due to the cost of a replacement. A survey
40 found that many Endodontists and General Dentists routinely access through
41 existing crowns [17], however, there is a lack of published data on the influence
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of accessing through existing crowns *versus* prior crown removal on the subsequent periapical healing and restorative complications.

The primary aim of this study was to radiographically determine the 2-year periapical healing associated with NSRCT completed through full coverage restorations in teeth with pre-operative periapical radiolucencies compared to those that had the existing restorations first removed (cuspal coverage and direct restorations) and subsequently restored with new full coverage restorations.

The null hypothesis was that there was no significant difference in periapical healing between teeth that had NSRCT completed through a pre-existing full coverage restoration, compared to teeth that received root canal treatment after restoration removal, coupled with subsequent placement of a new crown.

A secondary aim was to determine the complication rate (porcelain fracture and de-cementation) of accessing *through* existing extra-coronal restorations for up to 2 years later.

MATERIALS AND METHODS

Patient and tooth selection

Clinical records and radiographs of consecutive patients referred for NSRCT to manage periapical disease, and completed by a single operator (LF) at a Specialist Endodontic practice based in Poole, Dorset, United Kingdom, between July 2014 and December 2016, were identified. Only patients who had attended annual reviews of periapical healing for two years post-NSRCT were

1 included [2,18] but those with complete healing at one-year were not subjected
2 to further review.
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4 The selected root-treated teeth were divided into those where NSRCT had
5 been performed by accessing existing crowns, as well as those where existing
6 restorations (direct or indirect) were first removed to allow completion of NSRCT
7 plus placement of an immediate core and then subsequent restoration with a new
8 full coverage crown. Direct restorations commonly encountered and removed
9 before NSRCT included amalgam and a variety of tooth-coloured plastic filling
10 materials (resin-based, glass-ionomer-based, variety of interim restorative
11 material). Indirect restorations were either full-ceramic, full-metal or porcelain-
12 bonded-to-metal crowns.
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26 Patients with a medical history of diabetes or corticosteroids were
27 excluded. Teeth were excluded, if they had one or more of the following variables,
28 that were identified as potential confounding factors:
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- 33 1. radiographs of inadequate quality;
 - 34 2. history of previous surgical treatment;
 - 35 3. periodontal defects deeper than 5 mm [19];
 - 36 4. coronal cracks extending into the pulp chamber [20];
 - 37 5. pre- or intra-operative treatment mishaps (missed canals, perforations,
38 separated instruments) [21];
 - 39 6. absence of patency at the canal terminus [2];
 - 40 7. root fillings other than gutta-percha; or
 - 41 8. root fillings with voids or not extending within 2 mm of the radiographic
42 apex [22].
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Data collection

The following demographic and clinical data were extracted from the clinical notes: sex, age, tooth type, size of periapical lesion, presence of sinus tract and primary or secondary nature of root canal treatment (pre-operative), permanent core material (intra-operative), function as FPD abutment, cast restoration material and quality of coronal restoration (post-operative).

Clinical procedure

All patients were treated under local anaesthesia and rubber dam isolation. A dental operating microscope (Global Surgical Corporation, St Louis, MO) was used during the entire procedure. Existing large plastic restorations (amalgam or composite restorations) in the line of the access cavity were removed before NSRCT and any remaining restoration parts were removed before post-operative build-up. The decision to either keep or remove an existing cast cuspal-coverage restoration was reached in conjunction with the patient through informed consent. In general, patients were advised to have the crown removed if there was a poor marginal fit or tooth restorability was suspected to be questionable. However, in cases of suboptimal marginal adaptation but with an absence of gross marginal discrepancy and obvious caries, the patient made the final decision after being informed of the potential impact on treatment outcome. When accessing through a cuspal coverage restoration, a ball-ended diamond bur (Endo Access FG 2, Maillefer, Ballaigues, Switzerland) was used to remove porcelain and a tungsten carbide fissure bur (Jet Carbide FG 246, Kerr, Orange, CA, USA) to cut through metal with appropriate orientation, under water irrigation. Once in the pulp chamber the access was refined with an ultrasonic tip

(Start-X 3, Maillefer, Ballaigues, Switzerland) and the walls explored to determine the presence and size of restorations, caries and cracks. Following access, working length measurements were determined with a #10 or larger ReadySteel file (Dentsply Sirona, Ballaigues, Switzerland) using an electronic apex locator (EAL) (Root ZX II; J Morita, Irvine, CA) and the root canals were prepared with a reciprocating system (Wave One/Wave One Gold, Dentsply Sirona, Ballaigues, Switzerland). The EAL '0 reading' was taken as the apical limit of the preparation. A periapical radiograph was taken when apex locator readings were not stable or were discrepant with pre-operative radiographic estimated lengths. Master apical file size determination was based on triangulation of several factors: initial radiographically estimated canal size, evidence of debris on apical file-flutes during instrumentation and tactile feedback of apical file fit. The minimum apical preparation size was that of the Primary file of the reciprocating system (#25). Root canal filling materials were removed by using a combination of size 2 Gates-Glidden drills, hedstrom files or reciprocating instruments with or without the use of chloroform. The root canals were irrigated with 3% sodium hypochlorite (Teepol products, Orpington, UK) heated with a baby bottle warmer (Philips Avent, Philips, Guilford, UK) and a final rinse with 20% citric acid (Cerkamed, Stalowa Wola, Poland) delivered via a 27-G open-ended notched needle (Sherwood Medical, St. Louis, MO, USA). Both irrigant solutions were agitated manually with a matched gutta-percha cone, sonically (Endoactivator, Dentsply Sirona, Ballaigues, Switzerland), ultrasonically (Irrisafe, Acteon, Merignac, France) or by a combination of these. The canals were dried with matched paper points and obturated with matched gutta-percha cones (Dentsply Sirona, Ballaigues, Switzerland) and AH Plus Sealer (Dentsply Sirona, Ballaigues,

Switzerland) using a Schilder's warm vertical compaction technique. The tooth was then restored with dual-cure resin composite (Core-X, Dentsply Sirona, Ballaigues, Switzerland), glass ionomer cement (Fuji IX, GC America Inc., Alsip, IL, USA) or silver amalgam (Dispersalloy, Dentsply Sirona, Ballaigues, Switzerland). Intermediate Restorative Material (Dentsply Sirona, Ballaigues, Switzerland) orifice plugs were placed in conjunction with glass-ionomer cement and amalgam restorations. In general, accessed full-ceramic crowns were repaired with composite resin, whereas full-metal and porcelain-bonded-to-metal crowns were restored with either of the used core materials. Patients were referred back to the general dental practitioner after root canal treatment for the provision of cusp coverage restorations.

Radiographic examination

All periapical radiographs were taken with a paralleling technique using a positioning device (XCP holder; Rinn, Elgin, IL, USA). Conventional radiography with dental X-ray film (Insight; Eastman Kodak, Rochester, NY, USA) developed in an automated processing machine (Velopex Intra XE; Medivance Instruments, London, UK) was used for cases completed before March 2016. Digital radiography with an intraoral sensor (Planmeca ProScanner, Planmeca, Helsinki, Finland) was used for cases completed from March 2016 onwards. Conventional radiographs were placed on a light box and photographed in a standard manner using a DSLR camera in close-up default setting (Canon EOS 1100D, Canon Ltd, Uxbridge, UK) with a macro lens (Tamron AF 90mm f/2.8 Di SP A/M 1:1, Tamron Europe, Cologne, Germany) positioned in a horizontal camera mount (Alzo

Digital, Bethel, CT, USA). The images were then imported into the digital imaging software (Planmeca Romexis, Planmeca, Helsinki, Finland) and calibrated.

Radiographic evaluation of healing

An independent assessor (JR) examined the radiographs on a computer screen (iMac, Apple, Cupertino, CA, USA) in a dimly-lit room. A second observer (SM) assessed 30% of the sample to account for variation in radiographic assessment [2]. Examiners were able to visualize all radiographs of each case at the same time and could adjust brightness and contrast on the digital system software. The clinical crown was masked for radiographic periapical assessment. Both assessors were pre-calibrated by examining reference radiographs of the four categories of periapical healing outcomes:

1. *complete healing*, in the absence of apical abnormalities;
2. *incomplete healing*, when the pre-operative lesion had reduced in size without restitution of a normal periodontal ligament space;
3. *uncertain*, when it was not possible to make a decision on the progression of the periapical lesion;
4. *failure*, in case of persistent or worsening periapical radiolucency.

Radiographically-determined “success” was based on strict (only complete healing) or loose (both complete and incomplete periapical healing accepted as successful outcomes) criteria [2]. If a tooth had been extracted or had received further treatment because of endodontic problems, it was judged a failure. Multi-rooted teeth were assessed according to the root with the largest periapical radiolucency.

Evaluation of post-operative quality of the final restoration

Clinical examination of the restoration's marginal adaptation was carried out with a DG16/23 explorer (Hu-Friedy, Frankfurt Am Main, Germany) at a 90° angle to the restoration margin. The quality of the final extra-coronal restoration was dichotomized into: (i) *poor quality*; if there were clinical signs of recurrent caries, overhangs or a catch was detected with the explorer when run across the tooth/restoration interface, and radiographically if there were visible overhangs or gaps at the restoration margin; or (ii) *good quality*; if marginal fit was judged satisfactory both clinically and radiographically.

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opposite end of the spectrum, where the restoration may be better removed at

Compliance with ethical standards

Conflict of interest: The authors deny any conflicts of interest.

Ethical approval: This non-intervention retrospective cohort study did not require research ethics committee review according to the United Kingdom Health Research Authority (<http://www.hra-decisiontools.org.uk/research/>). This study was performed in accordance with the ethical standards as laid down in the 1964 Declaration of Helsinki and its later amendments or comparable ethical standards.

Informed consent: This study did not use identifiable details of patients and treatment data used was gathered anonymously.

REFERENCES

1. Quality guidelines for endodontic treatment: consensus report of the European Society of Endodontology (2006). *Int Endod J* 39 (12):921-930. doi:10.1111/j.1365-2591.2006.01180.x
2. Ng YL, Mann V, Gulabivala K (2011) A prospective study of the factors affecting outcomes of nonsurgical root canal treatment: part 1: periapical health. *Int Endod J* 44 (7):583-609. doi:10.1111/j.1365-2591.2011.01872.x
3. Gillen BM, Looney SW, Gu LS, Loushine BA, Weller RN, Loushine RJ, Pashley DH, Tay FR (2011) Impact of the quality of coronal restoration versus the quality of root canal fillings on success of root canal treatment: a systematic review and meta-analysis. *J Endod* 37 (7):895-902. doi:10.1016/j.joen.2011.04.002
4. Ng YL, Mann V, Gulabivala K (2010) Tooth survival following non-surgical root canal treatment: a systematic review of the literature. *Int Endod J* 43 (3):171-189. doi:10.1111/j.1365-2591.2009.01671.x
5. Burke FJ, Lucarotti PS (2009) Ten-year outcome of crowns placed within the General Dental Services in England and Wales. *J Dent* 37 (1):12-24. doi:10.1016/j.jdent.2008.03.017
6. Christensen GJ (2013) Too many crowns? *J Am Dent Assoc* 144 (10):1174-1176. doi:10.14219/jada.archive.2013.0037
7. Suzuki S, Cox CF, White KC (1994) Pulpal response after complete crown preparation, dentinal sealing, and provisional restoration. *Quintessence Int* 25 (7):477-485
8. Goodacre CJ, Bernal G, Rungcharassaeng K, Kan JY (2003) Clinical complications in fixed prosthodontics. *J Prosthet Dent* 90 (1):31-41. doi:10.1016/s0022-3913(03)00214-2
9. Pjetursson BE, Sailer I, Makarov NA, Zwahlen M, Thoma DS (2015) All-ceramic or metal-ceramic tooth-supported fixed dental prostheses (FDPs)? A

systematic review of the survival and complication rates. Part II: Multiple-unit FDPs. *Dent Mater* 31 (6):624-639. doi:10.1016/j.dental.2015.02.013

10. Sailer I, Makarov NA, Thoma DS, Zwahlen M, Pjetursson BE (2015) All-ceramic or metal-ceramic tooth-supported fixed dental prostheses (FDPs)? A systematic review of the survival and complication rates. Part I: Single crowns (SCs). *Dent Mater* 31 (6):603-623. doi:10.1016/j.dental.2015.02.011

11. Bergenholtz G, Nyman S (1984) Endodontic complications following periodontal and prosthetic treatment of patients with advanced periodontal disease. *J Periodontol* 55 (2):63-68. doi:10.1902/jop.1984.55.2.63

12. Kirkevang LL, Horsted-Bindslev P, Orstavik D, Wenzel A (2001) Frequency and distribution of endodontically treated teeth and apical periodontitis in an urban Danish population. *Int Endod J* 34 (3):198-205. doi:10.1046/j.1365-2591.2001.00370.x

13. Goldman M, Laosonthorn P, White RR (1992) Microleakage--full crowns and the dental pulp. *J Endod* 18 (10):473-475. doi:10.1016/s0099-2399(06)81345-2

14. Mulvey PG, Abbott PV (1996) The effect of endodontic access cavity preparation and subsequent restorative procedures on molar crown retention. *Aust Dent J* 41 (2):134-139. doi:10.1111/j.1834-7819.1996.tb05927.x

15. Whitworth JM, Walls AW, Wassell RW (2002) Crowns and extra-coronal restorations: endodontic considerations: the pulp, the root-treated tooth and the crown. *Br Dent J* 192 (6):315-320, 323-317. doi:10.1038/sj.bdj.4801365

16. Abbott PV (2004) Assessing restored teeth with pulp and periapical diseases for the presence of cracks, caries and marginal breakdown. *Aust Dent J* 49 (1):33-39; quiz 45. doi:10.1111/j.1834-7819.2004.tb00047.x

17. Trautmann G, Gutmann JL, Nunn ME, Witherspoon DE, Shulman JD (2000) Restoring teeth that are endodontically treated through existing crowns. Part I: Survey of pulpal status on access. *Quintessence Int* 31 (10):713-718

18. Bystrom A, Happonen RP, Sjogren U, Sundqvist G (1987) Healing of periapical lesions of pulpless teeth after endodontic treatment with controlled asepsis. *Endod Dent Traumatol* 3 (2):58-63. doi:10.1111/j.1600-9657.1987.tb00543.x

19. Ruiz XF, Duran-Sindreu F, Shemesh H, García Font M, Vallés M, Roig Cayón M, Olivieri JG (2017) Development of Periapical Lesions in Endodontically

Treated Teeth with and without Periodontal Involvement: A Retrospective Cohort Study. *J Endod* 43 (8):1246-1249. doi:10.1016/j.joen.2017.03.037

20. Sim IG, Lim TS, Krishnaswamy G, Chen NN (2016) Decision Making for Retention of Endodontically Treated Posterior Cracked Teeth: A 5-year Follow-up Study. *J Endod* 42 (2):225-229. doi:10.1016/j.joen.2015.11.011

21. Gorni FG, Gagliani MM (2004) The outcome of endodontic retreatment: a 2-yr follow-up. *J Endod* 30 (1):1-4. doi:10.1097/00004770-200401000-00001

22. Ng YL, Mann V, Rahbaran S, Lewsey J, Gulabivala K (2008) Outcome of primary root canal treatment: systematic review of the literature -- Part 2. Influence of clinical factors. *Int Endod J* 41 (1):6-31. doi:10.1111/j.1365-2591.2007.01323.x

23. Petrie A, Watson P (1999) *Statistics for veterinary and animal science*. 1st edn. Blackwell Science, Oxford, UK

24. He J, White RK, White CA, Schweitzer JL, Woodmansey KF (2017) Clinical and Patient-centered Outcomes of Nonsurgical Root Canal Retreatment in First Molars Using Contemporary Techniques. *J Endod* 43 (2):231-237. doi:10.1016/j.joen.2016.10.029

25. Hoskinson SE, Ng YL, Hoskinson AE, Moles DR, Gulabivala K (2002) A retrospective comparison of outcome of root canal treatment using two different protocols. *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 93 (6):705-715. doi:10.1067/moe.2001.122822

26. de Chevigny C, Dao TT, Basrani BR, Marquis V, Farzaneh M, Abitbol S, Friedman S (2008) Treatment outcome in endodontics: the Toronto study--phase 4: initial treatment. *J Endod* 34 (3):258-263. doi:10.1016/j.joen.2007.10.017

27. Chailertvanitkul P, Saunders WP, Saunders EM, MacKenzie D (1997) An evaluation of microbial coronal leakage in the restored pulp chamber of root-canal treated multirooted teeth. *Int Endod J* 30 (5):318-322. doi:10.1046/j.1365-2591.1997.00089.x

28. Ricucci D, Bergenholtz G (2003) Bacterial status in root-filled teeth exposed to the oral environment by loss of restoration and fracture or caries--a histobacteriological study of treated cases. *Int Endod J* 36 (11):787-802. doi:10.1046/j.1365-2591.2003.00721.x

29. Al-Maqtari AA, Lui JL (2010) Effect of aging on coronal microleakage in access cavities through metal ceramic crowns restored with resin composites. *J Prosthodont* 19 (5):347-356. doi:10.1111/j.1532-849X.2010.00593.x
30. Ng YL, Mann V, Gulabivala K (2008) Outcome of secondary root canal treatment: a systematic review of the literature. *Int Endod J* 41 (12):1026-1046. doi:10.1111/j.1365-2591.2008.01484.x
31. Bergenholtz G (2000) Evidence for bacterial causation of adverse pulpal responses in resin-based dental restorations. *Crit Rev Oral Biol Med* 11 (4):467-480. doi:10.1177/10454411000110040501
32. Whitworth JM, Myers PM, Smith J, Walls AW, McCabe JF (2005) Endodontic complications after plastic restorations in general practice. *Int Endod J* 38 (6):409-416. doi:10.1111/j.1365-2591.2005.00962.x
33. Browne RM, Tobias RS (1986) Microbial microleakage and pulpal inflammation: a review. *Endod Dent Traumatol* 2 (5):177-183. doi:10.1111/j.1600-9657.1986.tb00141.x
34. Moraschini V, Fai CK, Alto RM, Dos Santos GO (2015) Amalgam and resin composite longevity of posterior restorations: A systematic review and meta-analysis. *J Dent* 43 (9):1043-1050. doi:10.1016/j.jdent.2015.06.005
35. Mjör IA (2005) Clinical diagnosis of recurrent caries. *J Am Dent Assoc* 136 (10):1426-1433. doi:10.14219/jada.archive.2005.0057
36. Nagasiri R, Chitmongkolsuk S (2005) Long-term survival of endodontically treated molars without crown coverage: A retrospective cohort study. *J Prosthet Dent* 93:164-170. doi:10.1016/j.prosdent.2004.11.001
37. Gorman CM, Ray NJ, Burke FM (2016) The effect of endodontic access on all-ceramic crowns: A systematic review of in vitro studies. *J Dent* 53:22-29. doi:10.1016/j.jdent.2016.08.005
38. Gerogianni P, Lien W, Bompolaki D, Verrett R, Haney S, Mattie P, Johnson R (2019) Fracture Resistance of Pressed and Milled Lithium Disilicate Anterior Complete Coverage Restorations Following Endodontic Access Preparation. *J Prosthodont* 28 (2):163-170. doi:10.1111/jopr.12798
39. Bompolaki D, Kontogiorgos E, Wilson JB, Nagy WW (2015) Fracture resistance of lithium disilicate restorations after endodontic access preparation: An in vitro study. *J Prosthet Dent* 114 (4):580-586. doi:10.1016/j.prosdent.2015.04.013

40. Wood KC, Berzins DW, Luo Q, Thompson GA, Toth JM, Nagy WW (2006) Resistance to fracture of two all-ceramic crown materials following endodontic access. *J Prosthet Dent* 95 (1):33-41. doi:10.1016/j.prosdent.2005.11.003
41. Sutherland JK, Teplitsky PE, Moulding MB (1989) Endodontic access of all-ceramic crowns. *J Prosthet Dent* 61 (2):146-149. doi:10.1016/0022-3913(89)90362-4
42. McMullen AF, 3rd, Himel VT, Sarkar NK (1989) An in vitro study of the effect endodontic access preparation has upon the retention of porcelain fused to metal crowns of maxillary central incisors. *J Endod* 15 (4):154-156. doi:10.1016/s0099-2399(89)80252-3
43. Yu Y, Abbott P (1994) The effect of endodontic access cavity preparation and subsequent procedures on incisor crown retention. *Aust Dent J* 39:247-251. doi:10.1111/j.1834-7819.1994.tb04786.x
44. Skupien JA, Opdam N, Winnen R, Bronkhorst E, Kreulen C, Pereira-Cenci T, Huysmans MC (2013) A practice-based study on the survival of restored endodontically treated teeth. *J Endod* 39 (11):1335-1340. doi:10.1016/j.joen.2013.06.028
45. Wiegand A, Kanzow P (2020) Effect of Repairing Endodontic Access Cavities on Survival of Single Crowns and Retainer Restorations. *J Endod* 46 (3):376-382. doi:10.1016/j.joen.2019.11.012

Table 1 Reasons for exclusion of teeth

	Number of teeth
Inadequate image quality	4
Periodontal disease >5 mm	17
Coronal cracks into pulp chamber	18
NSRCT mishaps	17
Absence of apical patency	16
Root filling other than gutta-percha	4
Suboptimal post-operative root filling	4
Absence of cuspal coverage	107
Total	187

Table 2 Demographic and tooth-related characteristics by the treatment groups

	Crown retained	Crown removed	Restoration removed	P value
	106 teeth	57 teeth	82 teeth	
Sex				
Females (%)	60 (56.6%)	36 (63.2%)	33 (40.2%)	0.02
Males (%)	46 (43.4%)	21 (36.8%)	49 (59.8%)	
Age (years)	60±12	59±11	55±14	0.02
Tooth type (%)				
<i>Maxillary</i>				0.03
Incisors/canines	7 (6.6%)	9 (15.8%)	0 (0%)	
Premolars	15 (14.2%)	6 (10.5%)	10 (12.2%)	
Molars	35 (33.0%)	13 (22.8%)	27 (32.9%)	
<i>Mandibular</i>				
Incisors/canines	3 (2.8%)	3 (5.3%)	0 (0%)	
Premolars	6 (5.7%)	2 (3.5%)	5 (6.1%)	
Molars	40 (37.7%)	24 (42.1%)	40 (48.8%)	
Size of pre-operative periapical radiolucency (mm)	5.7±2.8 mm	6.5±3.0 mm	5.7±2.8 mm	0.2
Presence of pre-operative sinus tract (%)	32 (30.2%)	17 (29.8%)	14 (17.1%)	0.09
Previous treatment (%)	31 (29.3%)	37 (64.9%)	28 (34.2%)	<0.0001
Quality of final coverage restoration				
Good	62 (58.5%)	35 (61.4%)	48 (58.5%)	0.9
Poor	44 (41.5%)	22 (38.6%)	34 (41.5%)	
Crown material				
Porcelain bonded crown	75 (70.7%)	42 (73.7%)	50 (61%)	0.8
Full metal crown	13 (12.3%)	5 (8.8%)	22 (27%)	
Full ceramic crown	18 (17.0%)	10 (17.5%)	10 (12%)	
Post-RCT core material				
GIC	35 (33.0%)	3 (5.3%)	24 (29.3%)	<0.0001
Amalgam	20 (18.9%)	2 (3.5%)	6 (7.3%)	
Composite	51 (48.1%)	52 (91.2%)	52 (63.4%)	
Function as abutment at review	18 (17.0%)	3 (5.3%)	3 (3.7%)	0.004